DHBW Mosbach
Duale Hochschule
Baden-Württemberg

New Study

Registration of a project

Stand 11.4.2024

Studiengang New Study, DHBW Mosbach
Lohrtalweg 10, 74821 Mosbach

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☐ **Research Project IIa (Modul T3_2000, Work Integrated Project II, 3. Semester)**

☐ **Research Project IIb (Modul T3_2000, Work Integrated Project II, 4. Semester)**

☐ **Major Research Project II (Modul T3_2000, Work Integrated Project II, 3. - 4. Semester)**

☐ **Research Project III (Modul T3_3000, Work Integrated Project III, 5. Semester)**

☐ **Student Research Project (Modul T3_3101, 5. - 6. Semester)**

☒ **Bachelor Thesis (Modul T3_3300, 6. Semester)**

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Title of the thesis	**Comparison of a Classical RAG Pipeline and an Agent-Based Approach for Intelligent Knowledge Retrieval Across Enterprise Documentation Sources**
Motivation, problem statement, expected result	**In many enterprise environments, relevant knowledge is distributed across multiple systems such as Jira, Confluence, SharePoint, internal portals, and local documents. This fragmentation makes efficient information retrieval difficult, especially when team members need fast, reliable, and context-aware answers to operational or client-related questions. Manual search across these heterogeneous sources is time-consuming, error-prone, and often depends heavily on individual experience.** **A previous project addressed this challenge by extending the Ask Your Template (AYT) platform, a Retrieval-Augmented Generation (RAG) pipeline, with Jira and Confluence integration. In this setup, knowledge objects are ingested, normalized, chunked, vectorized, and made available for semantic retrieval. While this approach proved technically feasible and useful, it also revealed limitations regarding chunking strategy, source-specific retrieval quality, setup complexity, and long-term maintainability.** **At the same time, agent-based AI systems have recently gained relevance as an alternative paradigm for intelligent information access. Instead of relying only on retrieval over vectorized chunks, such systems can dynamically decide which source to query, how to decompose tasks, and how to combine retrieved information. This raises the question whether agent-based approaches may offer advantages over a classical RAG pipeline in practical enterprise settings.** **The objective of this thesis is therefore to design and implement a comparative evaluation of two approaches for intelligent cross-source knowledge retrieval: a classical RAG pipeline and an agent-based solution. Both approaches will be applied to the same use case and evaluated with regard to performance, cost, setup effort, and maintainability. Depending on implementation scope, retrieval quality and traceability may also be included as supplementary evaluation**

	criteria. **The expected result of the thesis is a structured comparison that identifies strengths, weaknesses, and trade-offs of both approaches in a realistic enterprise documentation scenario. The work should provide a practical decision basis for selecting an appropriate architecture for future intelligent knowledge systems in environments such as the TAP Rollout Team.**
Planned procedure	**he thesis will first establish the theoretical foundations of enterprise knowledge retrieval, RAG systems, and agent-based approaches, and define relevant evaluation criteria such as performance, cost, setup effort, and maintainability.** **Building on this, the existing RAG-based solution will be reviewed as a reference point, and an agent-based approach for the same retrieval problem will be conceptually designed and prototypically implemented. Both approaches will then be examined using representative enterprise use cases and compared based on the defined criteria.** **The exact implementation details and evaluation design may evolve during the course of the thesis. The overall aim is to derive a structured assessment of both approaches and to identify their respective strengths, weaknesses, and possible areas of application.**

Literature list (z.B. assigned literature, fundamental literature, literature pf preparation, advanced literature)	**Assigned literature / fundamental literature** **Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W. t., Rocktäschel, T., Riedel, S., and Kiela, D. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. Advances in Neural Information Processing Systems.** **Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., and Wang, H. (2024). *Retrieval-Augmented Generation for Large Language Models: A Survey*. arXiv.** **Mialon, G., Dessì, R., Lomeli, M., Ebrahimi, M., and others. (2023). *Augmented Language Models: a Survey*. arXiv.** **Russell, S., and Norvig, P. (2021). *Artificial Intelligence: A Modern Approach*. 4th ed. Pearson.** **Literature for preparation** **Atlassian. *Jira Software Documentation*. Official documentation.** **Atlassian. *Confluence Documentation*. Official documentation.** **Apache Lucene / vector search / embedding documentation depending on the chosen technical stack.** **Documentation of the AYT platform and internal project materials from the previous Jira and Confluence integration work.** **Advanced literature** **Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. (2023). *ReAct: Synergizing Reasoning and Acting in Language Models*. arXiv.** **Qin, Y., Zhang, S., Chen, D., and others. (2023). *Tool Learning with Foundation Models*. arXiv.** **Xi, Z., Chen, W., Guo, X., and others. (2023). *The Rise and Potential of Large Language Model Based Agents: A Survey*. arXiv.** **Chase, H. *LangChain Documentation* or equivalent framework documentation if used for the agent implementation.** **LanceDB, FAISS, or chosen vector database documentation depending on implementation.**
Langauge of elaboration	**German [ ] English [X]**
Date of creation of the topic message	**21/3/2026**

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